



AUTOMATION

The INDUSTRIES OF TOMORROW Will Be Run By The IIoT



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The Industrial Internet of Things enhances manufacturing and industrial processes. Let us find out how

The industry is a collection of three Ms: men, machines and materials. The primary function of the Internet of Things (IoT) is to integrate these effectively to meet industrial objectives. Several labs are, hence, researching various Internet technologies to apply to industrial situations.

The Industrial IoT (IIoT) promises to support a new era of smart industry, being a part of the IoT. The IoT is a network of intelligent computers, mobiles, gadgets and other objects that collect and share data. Collected data is sent to a central cloud-based service, where it is stored with other data and then shared with end users in a useful way. The IoT increases automation in homes, schools, stores and many other segments.

The IIoT has the potential to improve productivity across manufacturing sectors by integrating these with cloud computing, artificial intelligence (AI), deep learning and automation via the Internet. Hence, when the IoT is applied to the manufacturing industry, it is called IIoT.

Some key benefits of the IIoT are given below.

- Provides greater connectivity, efficiency and scalability
- Brings great savings in time and cost
- Improves safety and other operational efficiencies
- Improves the concept of Big Data

- Has openness and inter-operability, which means that it can connect all machines of an industry having different protocols in an effective manner

- Enhances supply chain efforts
- Provides green practices
- Provides high-quality control

A number of companies are implementing IIoT technologies by taking maximum advantage of intelligent and connected devices in their factories. One of the main characteristics of the industry is the presence of machines. Thus, any sector that makes use of machines can be termed as industry. Automobile, oil and gas, electrical and power, agriculture and hospitals are typical examples of such industries.

The IIoT connects processes from the factory floor to executive offices. Business leaders today are extensively using IIoT, as it helps them get a full view of how their enterprise is functioning. This helps them make better decisions.

In factories, devices and machines use different protocols and have different architectures. These might be sold by different vendors. Connecting these devices and machines, and providing security is the greatest challenge for the IIoT today.

Ignition is an excellent solution for this. It works on a cross platform, and is built on open source and standard IT technologies.